An Average-Cost Dynamic Reputation Model with Hidden Product Choice

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We study an infinite-horizon reputation problem in which a seller privately chooses between a low-cost product and a high-quality product while users learn only from realized binary outcomes. The user treats the seller as a single Bernoulli arm and uses Thompson sampling to decide whether to engage with the seller or with an outside option. The seller chooses products to maximize long-run average profit per calendar period. The resulting control problem is an average-reward Markov decision process on the user's posterior sufficient statistics. The formulation characterizes the seller's optimal hidden product choice through the relative value of a success versus a failure.

Key words: dynamic reputation; hidden actions; average-cost Markov decision processes; Thompson sampling

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1. Model Formulation

Consider an infinite-horizon discrete-time setting, indexed by $t \in \mathbb{N}^+ = \{1, 2, \ldots\}$. The environment consists of a single user and two sellers, denoted by A and B. At each period t, the user selects a seller, denoted by $y_t \in \{A, B\}$. The user's realized utility in period t is a binary random variable $u_t \in \{0, 1\}$.

Seller B offers a transparent outside option. If the user selects Seller B, then the realized utility is drawn from a Bernoulli distribution with known success probability $p_0 \in (0, 1)$:

$$\mathbb{P}(u_t = 1 \mid y_t = B) = p_0. \tag{1}$$

Seller A has access to two products, indexed by $\mathcal{X} = \{1, 2\}$. When the user selects Seller A, the seller privately chooses a product $x_t \in \mathcal{X}$. Product $i \in \mathcal{X}$ generates a successful outcome for the user with probability p_i and imposes cost c_i on Seller A:

$$\mathbb{P}(u_t = 1 \mid y_t = A, x_t = i) = p_i. \tag{2}$$

We assume

$$0 < p_1 < p_2 \leq 1, \qquad c_1 < c_2, \tag{3}$$

so product 2 is more likely to satisfy the user but is also more costly. Whenever Seller A is chosen, it receives a fixed revenue $R > 0$. Seller A's period profit is

$$\pi_t^A = \begin{cases} R - c_{x_t} & \text{if } y_t = A, \\ 0 & \text{if } y_t = B. \end{cases} \tag{4}$$

2. Information Structure and Learning Dynamics

The central feature of the model is that Seller A's product choice is hidden from the user. The product affects the distribution of realized utility, but the user does not observe which product generated the realized outcome.

ASSUMPTION 1 **(User's Misspecified Belief)**. *The user observes the realized utility u_t after choosing Seller A, but does not observe the product $x_t \in \mathcal{X}$ assigned by Seller A. The user therefore treats Seller A as a single product characterized by an unknown Bernoulli success probability $\theta \in [0,1]$.*

The user learns about θ through Bayesian updating. Let S_t and F_t denote the cumulative number of successes and failures generated by Seller A up to the beginning of period t:

$$S_t = \sum_{\tau=1}^{t-1} \mathbb{1}\{y_\tau = A, u_\tau = 1\}, \tag{5}$$

$$F_t = \sum_{\tau=1}^{t-1} \mathbb{1}\{y_\tau = A, u_\tau = 0\}. \tag{6}$$

The user starts from the uniform prior $\theta \sim \mathrm{Beta}(1,1)$. Given state (S_t, F_t), the posterior belief is

$$\theta \mid S_t, F_t \sim \mathrm{Beta}(1+S_t, 1+F_t). \tag{7}$$

The user follows a Thompson sampling heuristic. At the beginning of period t, the user samples

$$\tilde{\theta}_t \sim \mathrm{Beta}(1+S_t, 1+F_t). \tag{8}$$

Because the outside option has known success probability p_0, the user chooses Seller A if and only if the sampled index exceeds p_0:

$$y_t = \begin{cases} A & \text{if } \tilde{\theta}_t \geq p_0, \\ B & \text{if } \tilde{\theta}_t < p_0. \end{cases} \tag{9}$$

Seller A observes the user's choices and realized outcomes, and can therefore track the sufficient statistics (S_t, F_t). For a state (S, F), define Seller A's demand probability by

$$\rho(S,F) \triangleq \mathbb{P}(\tilde{\theta}_t \geq p_0 \mid S_t = S, F_t = F) = \int_{p_0}^{1} \frac{\theta^S (1-\theta)^F}{B(1+S, 1+F)}\, d\theta, \tag{10}$$

where $B(\cdot,\cdot)$ is the Beta function.

3. Seller A's Average-Cost Optimization Problem

Seller A faces a Markov decision problem with state $\boldsymbol{s}_t = (S_t, F_t) \in \mathbb{N}_0 \times \mathbb{N}_0$. A stationary policy is a mapping $\pi_A : \mathbb{N}_0 \times \mathbb{N}_0 \to \mathcal{X}$ that assigns a hidden product to each posterior state. The seller's objective is to maximize long-run average profit per calendar period:

$$\eta^* = \sup_{\pi_A} \liminf_{T\to\infty} \frac{1}{T} \mathbb{E}_{\pi_A} \left[\sum_{t=1}^{T} \mathbb{1}\{y_t = A\}(R - c_{x_t}) \right]. \tag{11}$$

Equivalently, the same problem can be written as an average-cost problem by minimizing the negative of Seller A's profit.

Let η denote the optimal average profit and let $h(S,F)$ denote the relative value, or bias, of state (S,F). When the user chooses Seller B, calendar time advances but Seller A's reputation state remains (S,F). When the user chooses Seller A and product x is used, the next state is $(S+1,F)$ after a success and $(S,F+1)$ after a failure. The average-reward optimality equation is therefore

$$\begin{aligned} \eta + h(S,F) = &[1-\rho(S,F)]h(S,F) \\ &+ \rho(S,F) \max_{x\in\{1,2\}} \{R - c_x + p_x h(S+1,F) + (1-p_x)h(S,F+1)\}. \end{aligned} \tag{12}$$

For each product x, define the product-specific average-reward action value

$$Q_x(S,F) \triangleq R - c_x + p_x h(S+1,F) + (1-p_x)h(S,F+1). \tag{13}$$

The demand probability $\rho(S,F)$ determines how often Seller A has the opportunity to act, while the product choice at a fixed state is governed by the relative values embedded in $Q_x(S,F)$. Seller A chooses product 2 whenever

$$Q_2(S,F) \geq Q_1(S,F), \tag{14}$$

or equivalently whenever

$$h(S+1,F) - h(S,F+1) \geq \frac{c_2 - c_1}{p_2 - p_1}. \tag{15}$$

Thus, the high-quality product is optimal precisely when the relative value of a success rather than a failure is large enough to justify the additional product cost.